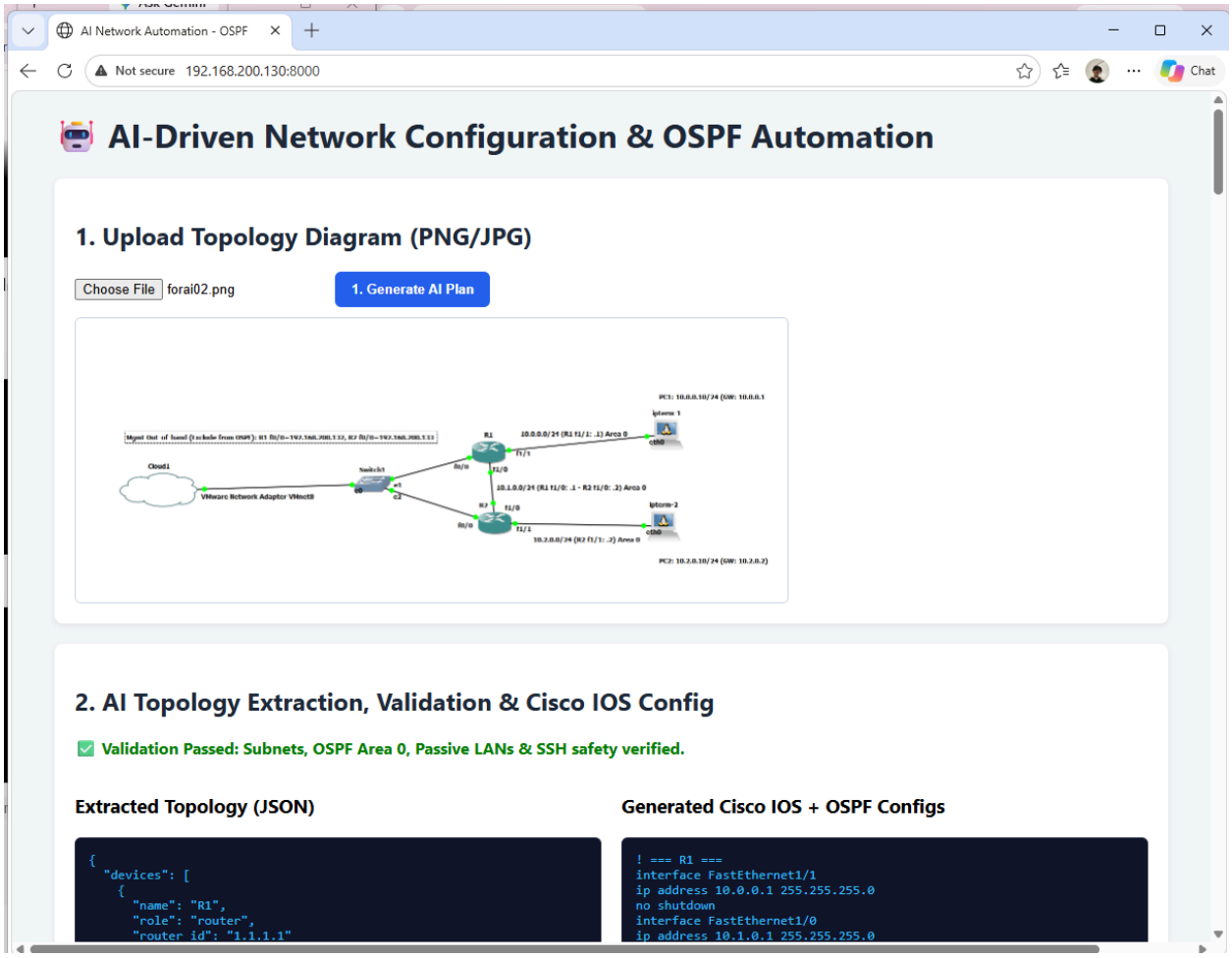
(AI Analysis)


AI Network Automation - OSPF
Not secure 192.168.200.130:8000
Chat
AI-Driven Network Configuration & OSPF Automation
1. Upload Topology Diagram (PNG/JPG)
Choose File forai02.png
1. Generate AI Plan
2. AI Topology Extraction, Validation & Cisco IOS Config
Validation Passed: Subnets, OSPF Area 0, Passive LANs & SSH safety verified.
Extracted Topology (JSON)
{
"devices": [
{
"name": "R1",
"role": "router",
"router id": "1.1.1.1"
Generated Cisco IOS + OSPF Configs
! === R1 ===
interface FastEthernet1/1
ip address 10.0.0.1 255.255.255.0
no shutdown
interface FastEthernet1/0
ip address 10.1.0.1 255.255.255.0

Job ID: job-5592da3d | **Router:** R1 | SUCCESS | Updated: 2026-11-12T07:43:46.722698

Paramiko Deployment Output:

```
Warning: Permanently added '192.168.200.132' (RSA) to the list of known hosts.

R1#terminal length 0
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#configure terminal
                ^
% Invalid input detected at '^' marker.

R1(config)#interface FastEthernet1/1
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 10.1.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#passive-interface FastEthernet1/1
R1(config-router)#network 10.0.0.0 0.0.0.255 area 0
R1(config-router)#network 10.1.0.0 0.0.0.255 area 0
R1(config-router)#end
R1#end
   ^
% Invalid input detected at '^' marker.

R1#write memory
Building configuration...
[OK]
R1#
R1#exit
Connection to 192.168.200.132 closed by remote host.

Connection to 192.168.200.132 closed.
```

OSPF Verification (Neighbor / Routes / Protocols):

```
Warning: Permanently added '192.168.200.132' (RSA) to the list of known hosts.

R1#terminal length 0
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:34    10.1.0.2        FastEthernet1/0
R1#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 172.28.152.1 to network 0.0.0.0

      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 10.1.0.2, 00:02:22, FastEthernet1/0
     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O        10.2.0.0/24 [110/2] via 10.1.0.2, 00:02:22, FastEthernet1/0
R1#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.255 area 0
    10.1.0.0 0.0.0.255 area 0
  Routing on Interfaces Configured Explicitly (Area 0):
    Loopback0
    FastEthernet1/0
    FastEthernet1/1
  Passive Interface(s):
    FastEthernet1/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    2.2.2.2              110      00:02:22
  Distance: (default is 110)

R1#exit
Connection to 192.168.200.132 closed.
```

Job ID: job-3c871a2d | **Router:** R2 | SUCCESS | Updated: 2026-11-12T07:44:28.214940

Paramiko Deployment Output:

```
Warning: Permanently added '192.168.200.133' (RSA) to the list of known hosts.

R2#terminal length 0
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#configure terminal
                ^
% Invalid input detected at '^' marker.

R2(config)#interface FastEthernet1/0
R2(config-if)#ip address 10.1.0.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface FastEthernet1/1
R2(config-if)#ip address 10.2.0.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#passive-interface FastEthernet1/1
R2(config-router)#network 10.1.0.0 0.0.0.255 area 0
R2(config-router)#network 10.2.0.0 0.0.0.255 area 0
R2(config-router)#end
R2#end
   ^
% Invalid input detected at '^' marker.

R2#write memory
Building configuration...
[OK]
R2#
R2#exit
Connection to 192.168.200.133 closed by remote host.

Connection to 192.168.200.133 closed.
```

OSPF Verification (Neighbor / Routes / Protocols):

```
Warning: Permanently added '192.168.200.133' (RSA) to the list of known hosts.

R2#terminal length 0
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:35    10.1.0.1        FastEthernet1/0
R2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 10.1.0.1, 00:03:03, FastEthernet1/0
     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O        10.0.0.0/24 [110/2] via 10.1.0.1, 00:03:03, FastEthernet1/0
R2#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.1.0.0 0.0.0.255 area 0
    10.2.0.0 0.0.0.255 area 0
  Routing on Interfaces Configured Explicitly (Area 0):
    Loopback0
    FastEthernet1/1
    FastEthernet1/0
  Passive Interface(s):
    FastEthernet1/1
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:03:04
  Distance: (default is 110)

R2#exit
Connection to 192.168.200.133 closed.
```

การตรวจสอบ OSPF Neighbors และ OSPF Routes

```
R1#
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:36    10.1.0.2        FastEthernet1/0
R1#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 172.28.152.1 to network 0.0.0.0

      2.0.0.0/32 is subnetted, 1 subnets
O        2.2.2.2 [110/2] via 10.1.0.2, 00:39:28, FastEthernet1/0
      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O        10.2.0.0/24 [110/2] via 10.1.0.2, 00:39:28, FastEthernet1/0
R1#
```

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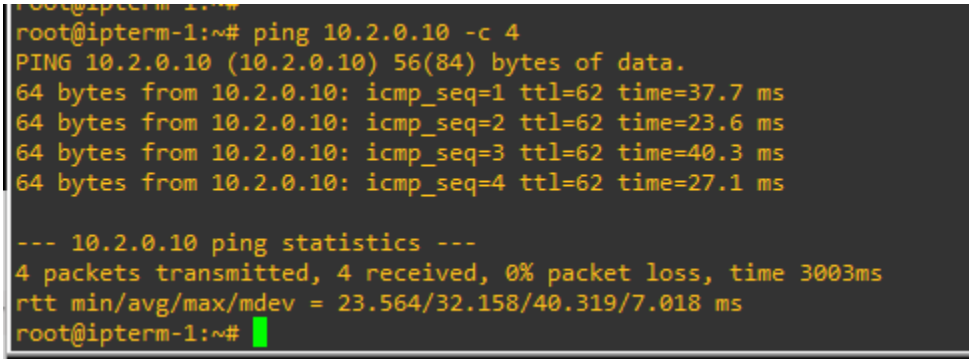
```
R2#
R2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
1.1.1.1           1   FULL/DR         00:00:39    10.1.0.1        FastEthernet1/0
R2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
O        1.1.1.1 [110/2] via 10.1.0.1, 00:40:03, FastEthernet1/0
      10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O        10.0.0.0/24 [110/2] via 10.1.0.1, 00:40:03, FastEthernet1/0
R2#
```

End-to-End Connectivity (PC1 ping PC2 สำเร็จ)



1. System Architecture Overview

*The **AI-Driven Network Automation System** is designed using a decoupled, event-driven architecture with a strict Human-in-the-Loop approval workflow to ensure operational safety:*

- ***Web Frontend & Backend (FastAPI):** Accepts network topology diagram uploads (PNG/JPG), triggers AI analysis, performs automated pre-deployment validation (checking subnets, OSPF settings, and SSH safety), and provides a review interface requiring explicit administrator approval before dispatch.*
- ***AI Diagram Analysis (Multimodal LLM):** Extracts network devices, interfaces, IP addressing plans, and OSPFv2 (Process 1, Area 0) requirements into structured JSON, generating Cisco IOS configurations with passive-interface enabled on LAN links while excluding out-of-band management interfaces (FastEthernet0/0).*
- ***Message Broker (Apache Kafka):** Upon administrator approval, the backend publishes validated deployment jobs with unique job IDs to the network-config-jobs topic for asynchronous distribution.*
- ***Execution Worker & SSH Automation (Python + Paramiko):** A dedicated Python worker consumes jobs from Kafka, re-verifies approval status with the backend, and connects to R1 and R2 via SSH using Paramiko (loading credentials securely from environment variables without hardcoding). It deploys the configuration and automatically executes verification commands.*

2. Test Results & Verification

End-to-end testing in the GNS3 lab environment confirmed that all automation and routing requirements were met:

1. ***AI Extraction & Validation:** The AI model accurately extracted the topology, assigned Router IDs (1.1.1.1 for R1 and 2.2.2.2 for R2), and passed all automated validation checks.*
2. ***Automated Deployment & OSPF Adjacency:** The worker successfully deployed configurations via SSH (SUCCESS status). R1 and R2 established OSPFv2 neighbor adjacency over FastEthernet1/0 (10.1.0.0/24), reaching the **FULL** state on both routers.*
3. ***Dynamic Routing & Passive Interfaces:** R1 dynamically learned the remote LAN 10.2.0.0/24 via OSPF (O), while R2 learned 10.0.0.0/24 via OSPF (O). LAN-facing interfaces (FastEthernet1/1) were verified as passive.*
4. ***End-to-End Connectivity:** ICMP ping tests from PC1 (10.0.0.10) to PC2 (10.2.0.10) succeeded with **0% packet loss** (4 packets transmitted, 4 received), confirming full end-to-end routing via OSPF.*

3. Repository & Submission Links

- ***Source Code & Documentation:** https://github.com/6631501054-Teradon/ai-netauto-ospf (Contains application source code, Docker Compose configuration, .env.example template, and README.md setup instructions with all secrets excluded).*